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[About the Embedded Signal Analyzer Toolkit](#) : VSA Help

VSA Help

Keysight Technologies 89600 VSA Software comes with its own help system which includes general VSA information and language reference for the COM API.

Use the following button to launch VSA help:

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Use the following button to launch VSA .COM API help:

After VSA installation, Keysight Technologies 89600 VSA Software help is located at:

C:\Program Files\Agilent\89600 Software 17.2\89600 VSA Software\Help

To launch VSA help, double-click VSA Help.bat in this folder.

To launch .NET COM API help, double-click .NET API Reference.bat in this folder.

You can also launch VSA help from the Start menu:

Start>All Programs>Agilent 89600 Software 17.2>Help

		
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About the Embedded Signal Analyzer Toolkit : ESA Toolkit Installation

ESA Toolkit Installation

To use the ESA Toolkit, you must do the following:

- | | |
|----|---|
| 1. | Install ESA Toolkit. (The ESA installation procedure also installs Keysight Technologies 89600B VSA 17.22.) |
| 2. | Set up the license server and licenses for VSA. |
| 3. | Install the Teradyne Signal Library using downloads from eKnowledge. |

For more information, see [ESA V4.5 Toolkit Installation Instructions](#) in eKnowledge.

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[About the Embedded Signal Analyzer Toolkit](#) : ESA Toolkit Software Licensing Requirements

ESA Toolkit Software Licensing Requirements

To run demodulation using ESA, you need a VSA license from [Keysight Technologies](#).
For more information, see [IG-XL Licensing](#) in the IG-XL Administration and Licensing section.

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About the Embedded Signal Analyzer Toolkit : ESA Toolkit Software Licensing Requirements : Running a Test Program Off-Line Using ESA Without a VSA License

Running a Test Program Off-Line Using ESA Without a VSA License
Given the current IG-XL and 89600 VSA Software licensing options, you may or may not have all the required licensing to successfully use a test program off-line with the ESA Toolkit. This section describes the options available to you.

Background
IG-XL enables all Teradyne licensed hardware options when run off-line or when running checkers. All hardware option licenses are defined and issued by Teradyne.
Teradyne includes third party software with IG-XL (such as SEDana and WaveWizard), which includes limited functionality through licenses. You can purchase licenses with additional features from third parties.
The ESA Modulation Toolkit is a licensed software option in the IG-XL environment. ESA demodulation uses VSA software, which is licensed by Keysight Technologies:

- Each enabled seat of ESA (VSA) provided by Teradyne requires a license purchased by Teradyne from Keysight Technologies.
- Each license is node locked to a specific tester or off-line PC.
- The ESA Modulation Toolkit provides a means of debugging ESA-based demodulation off-line using an off-line ESA (VSA) license

ESA and VSA Issues
Suppose that you have an ESA based test program but no off-line ESA (VSA) license. Can you load and validate the program off-line? Can you debug the non-ESA portion of the program? The

following table lists possible licensing issues, as well as other requirements for using ESA and VSA for test programs.

ESA and VSA Issues	
Issue	Effect
ESA Toolkit not installed	ESA-specific VBT language (.Demod and .DemodDSP) in the program causes validation errors due to ESA code contained in DSP procedures which are “compiled” during validation.
ESA Toolkit installed, no VSA license	Program successfully validates, but attempt to start VSA using ESA-based demodulation language (.DemodDSP) or VSA debuggers (.Demod.Debuggers) returns run-time errors. You may need to manipulate program to not use the VSA.
Using new ESA features on older version of IG-XL	IG-XL returns an error. You must have a valid license agreement to get the latest version of IG-XL and ESA. Optionally, you can edit the program to remove or limit use of language associated with a new feature.
Teradyne Signal Library file missing	Test program does not compile. Validation errors indicate missing libraries for the language used in the test program.

Options for Using ESA

The following are options available to you for running test programs that use ESA off-line:

- Purchase an off-line ESA Modulation Toolkit license for a specific PC.

This provides access to full IG-XL and ESA capabilities on one machine.

- Purchase an off-line ESA Modulation Toolkit license for a shared PC.

- Provides access to full capabilities of IG-XL and ESA through a remote machine using such utilities as Microsoft Remote Desktop and VNC

- Availability conflicts may arise if use is high by many users

- Install ESA Modulation Toolkit on a local machine without an off-line ESA (VSA) license

- Programs validate and run without error unless you attempt to access VSA instances (using .DemodDSP or VSA debuggers)
- You can use standard IG-XL techniques to control the flow so that VSA access is not attempted. For example, you can use different flows on the Job sheet, enable words on a Flow sheet, use global variables, and use conditional VBT to control lower level code execution. You can also use tools such as ASCIIUtils.xla to assist in loading variations of sheets and modules depending on the execution environment.

- Do not load ESA Toolkit on your local machine

- You must remove any ESA-based VBT language (.Demod and .DemodDSP) from the test program to successfully validate the job.
- You can design the program to isolate ESA code in specific VB modules and then use tools such as ASCIIUtils.xla to assist in loading different variations of sheets and modules depending on the execution environment.

Additional points to consider:

- How difficult is installing the ESA Toolkit without a VSA license?
 - Customers with a valid ESA license have access to the media for installation.
 - Install time for a Teradyne Signal Library is about 5 minutes on a typical PC.
- How difficult is manipulating program flow for ESA?
 - Typically, excluding ESA based code execution is done using “test instance granularity” on a Flow sheet. This allows you to execute or not execute ESA tests in the flow.
 - Adding or modifying a few enable words on a Flow sheet can take under one minute.
 - Creating a Job sheet (if one does not already exist for the program), copying a Flow sheet to create a “non-ESA” flow, and making edits on that Flow sheet can typically be done in 5-

10 minutes and is reusable for that job into the future.

- If your program does not allow control of flow at the test instance level, then you must use lower level edits such as If-Then-Else within instances. This work is job-specific, but the time required is still probably less than an hour and would be reusable.



ESAToolkitDefTopic

	
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About the Embedded Signal Analyzer Toolkit

About the Embedded Signal Analyzer Toolkit

The Embedded Signal Analyzer (ESA) Toolkit is a software option available for use with IG-XL on UltraFLEX. The ESA Toolkit provides integrated tools for developing and deploying RF device tests which require modulated signal generation and measurements using standard wireless communication protocols, such as GSM/GPRS, EDGE, WCDMA, Bluetooth, LTE, and 802.11. The ESA tools are used primarily with the UltraWave 12G and UltraWave24 MW Source and Receiver instruments, but the waveform sample data in signal definitions can also be used in certain applications with other instruments, such as the BBAC.

The ESA Toolkit provides the following advantages during development of RF tests requiring standards-based signal modulation or demodulation:

- Simple and fast test program development using Teradyne provided standards-based modulated signals and vector signal analyzer setups. Test engineers do not have to be experts in modulation standards.
- High multisite test throughput using the IG-XL DSP environment running on the two dual-core DSP computers available on UltraFLEX
- Correlation of test results with bench equipment and design results. Modulated source signals and the demodulation measurement results are based on industry standard EDA tools and measurements.

The ESA Toolkit contains the following components:

- Teradyne Signal Library

The Teradyne signal library contains hundreds of modulation (source) signal definitions and demodulation (measurement or capture) signal definitions. Signal definitions are supplied in the Teradyne Signal Definition (TSD) format, an XML-based file format. The signal definitions in TSD files are available in eKnowledge in groups that are based on wireless applications and standards.

- [Vector Signal Analyzer \(VSA\)](#)

The embedded run-time vector signal analyzer is Keysight Technologies 89600B VSA Software. This software allows streaming of waveform data from an UltraWave 12G or UltraWave24 MW Receiver instrument to one or more VSA demodulator processes acting as compute engines. The VSA processes operate in the background in the UltraFLEX DSP environment for high multisite throughput.

During test program debugging, up to two debugger VSAs with graphical user interfaces (GUIs) are available for interactive display and control.

- [VBT Language: Demod object, DemodDSP object, DemodDSPMacro object, Signals Interface for Instruments](#)

The Demod, DemodDSP, and DemodDSPMacro VBT objects provide properties and methods for creating, configuring, and communicating with one or more VSA demodulator processes, including the debugger (visible, with GUI) processes. All communications with VSA processes are managed by the Demod service.

Signal definitions are easily applied to the UltraWave instruments and the VSAs in a test program using the VBT Signals interface for UltraWave 12G and UltraWave24 instruments.

IG-XL includes a standalone tool that is not part of the ESA Toolkit option but is essential for working with the Modulation Signal Library and TSD files:

- [Signal Definition Explorer \(SDE\)](#)

SDE is a TSD file browser and editor. SDE includes functions to manage libraries of TSD files, manage the waveform data in TSD files, and move or copy TSD files to test program development folders outside of the TSD libraries.

This section includes the following topics:

- [VSA Help](#)

- [ESA Toolkit Installation](#)

- [ESA Toolkit Software Licensing Requirements](#)

ESA Toolkit information includes the following sections:

- [Using the ESA Toolkit](#)

Demodulation Software Support

The Embedded Signal Analyzer (ESA) Toolkit supports a number of demodulation standards. Use of ESA for these standards requires a VSA license.

The ESA Toolkit supports the following standards:

- [GSM/GPRS](#)

- [EDGE](#)

- [WCDMA](#)

- [WCDMA - HSDPA](#)

- [WCDMA - HSUPA](#)

- [TD-SCDMA](#)

- [CDMA ONE \(IS95\)](#)

- [CDMA2000](#)

- [1xEV-DO](#)

- [Bluetooth 1.1/1.2](#)

- [Bluetooth 2.0/2.1 + EDR](#)

- [Bluetooth 3.0](#)

•	ZigBee (802.15.4)
•	802.11b
•	802.11a/g
•	802.11n
•	802.11ac
•	802.11p
•	WiMax-e (Mobile, 802.16e)
•	WiMax-d (Fixed, 802.16d)
•	WiBro
•	DECT
•	FM
•	DVB-H (stimulus only)
•	DVB-T (stimulus only)
•	3 GPP LTE 8.9 FDD
•	LTE-Advanced

Related Topics

•	ESA Toolkit Software Licensing Requirements
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	About the Signal Definition Explorer
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Using the ESA Toolkit

Using the ESA Toolkit

The ESA Toolkit provides tools for developing tests for RF devices that require standards-based modulated input signals or produce standards-based modulated output signals, or both. This guide describes the components of the ESA Toolkit, how they operate, and how to use them. This section includes the following topics:

- [Overview of the ESA Toolkit](#)
- [Demod Operation on Tester Computers](#)
- [Introduction to VSA](#)
- [Sourcing and Analyzing Modulated Signals](#)
- [Debugging Demodulation Tests](#)

Related Topics

- [Using the Signal Definition Explorer](#)
- [Demod](#) (VBT object)
- [DemodDSP](#) (VBT object)

	• DemodDSPMacro (VBT object)
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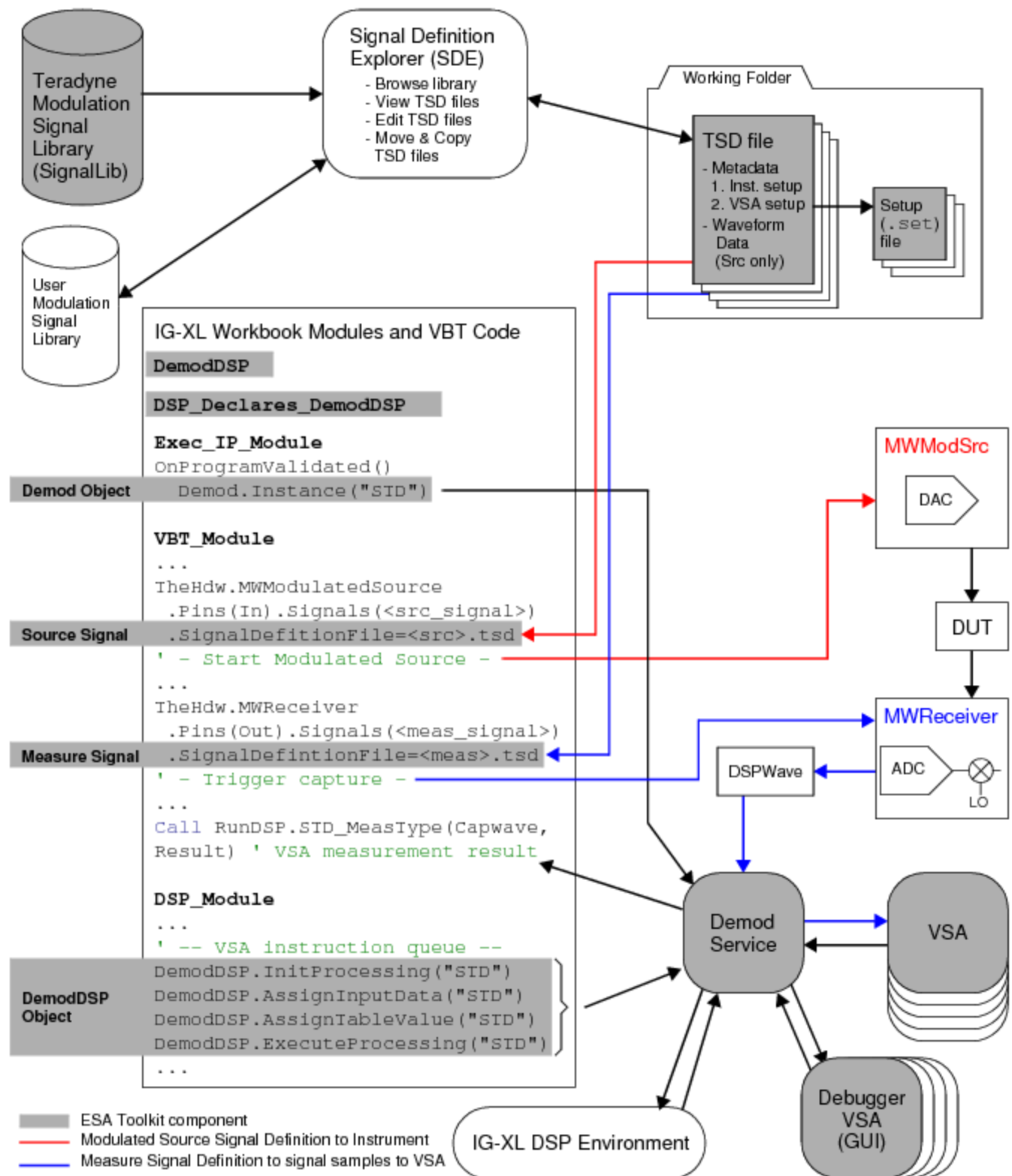


[Using the ESA Toolkit](#) : Overview of the ESA Toolkit

Overview of the ESA Toolkit

In addition to describing ESA Toolkit components, this overview describes, at a high level, how ESA components operate and some of the ways that you use them during modulation test development.

The figure below is a diagram of ESA Toolkit components in the context of IG-XL and UltraFLEX hardware. The diagram is a conceptual view of ESA components, with VBT objects and instrument signal interfaces represented as code fragments. ESA Toolkit components and code are shaded.



ESA Toolkit Components On IG-XL/UltraFLEX with UltraWave 12G or UltraWave24

Signal Libraries and Signal Definitions

The Teradyne Modulation Signal Library that is installed on your workstation is shown in the upper left corner. This library is also referred to as the Teradyne Signal Library or SignalLib (the name of the folder it is installed in) or the Teradyne TSD library. This library contains signal definitions for wireless communication standards. Each signal definition contains instrument and test specific parameter settings and waveform data (for source signals only) for the stimulus signal or capture

and measurement of a specific standards-based test. Each definition also includes the setup for the run-time analyzer used to demodulate a source or measure signal.

Just below the SignalLib, the diagram shows the smaller User Modulation Signal Library. It is also referred to as the User Signal Library, or user TSD library. This library is available for you to store and organize signal definitions that you create by modifying signal definitions supplied by Teradyne or by importing existing waveform data.

TSD Files

Each signal definition is implemented as a Teradyne Signal Definition (TSD) file. In the diagram, a set of TSD files is shown in the upper right corner, located in a working folder accessible to your test program. Typically, most TSD files originate in the SignalLib.

Although the details are not shown, the Signal libraries are folders of TSD files. In the case of the Teradyne SignalLib, TSD files are organized under groups of wireless products, such as 3G cellular, connectivity, and cordless phone. Under each group is a folder for each specific wireless protocol that contains TSD files for specific tests or test development purposes. In some cases, you can use TSD files from the user TSD library that were developed by you or other test engineers. (An IG-XL tool for browsing and managing TSD files and signal libraries, Signal Definition Explorer, is described below.)

No matter how the TSD files were created, you copy the ones your test plan requires to a folder accessible to your IG-XL test program workbook. You modify the TSD files used by the test program in the working folder.

Signal definitions, as implemented in TSD files, are stored in an XML-based file format. TSD files may contain signal waveform data, and metadata. Metadata (data about data) describes the characteristics of the waveform data needed to realize it as a physical signal at the instrument, including the instrument setup for the test. Setup parameters can be loaded into UltraWave instruments using their VBT Signals interfaces.

Source TSD files have waveform data and metadata. Measurement (capture) TSD files do not have waveform data. Metadata includes references to separate files in the same directory that contain setup or other data required by the demodulator software. The diagram shows setup files associated with TSD files in the working directory. TSD files may contain multiple, unique streams of waveform data.

SDE

At top center in the diagram is the Signal Definition Explorer (SDE), a tool with a graphical user interface. (SDE is installed with IG-XL and not as part of the ESA Toolkit option.) You use SDE to find, view, choose (each file has a description), and edit TSD files, and to manage files and signal libraries. A key function of SDE is to automatically copy the setup file and any utility files referenced by a TSD file that you copy to another folder. Another key function is to verify that a TSD file has the correct file syntax. You use SDE to copy TSD files to one or more working directories, as shown in the diagram.

VBT Language, DSP Library, and the Demod Service

The VBT Demod, DemodDSP, and DemodDSPMacro objects and the Signals interfaces for the RF instruments are in the IG-XL workbook in the left center and lower left of the diagram. These VBT language elements are illustrated as VBT code fragments, located in specific VB modules in the test program workbook. The Demod Service provides the central communication and supervision functions that allow the demodulator software (VSA) to operate with IG-XL, and integrates the operation of TSD files, the VBT objects and interfaces, and the demodulator software with IG-XL, including the UltraFLEX instrument drivers and the DSP environment.

The Demod service is available for use after the ESA Toolkit is installed and the declares file for the DemodDSP library is imported into the workbook. After the program is validated, IG-XL automatically adds two modules to the workbook, named DemodDSP and DSP_Declares_DemodDSP. Do not edit these modules.

The features of an IG-XL workbook using the ESA Toolkit option are highlighted with shading in the diagram:

- DSP_Declares and DSP_Declares_DSP modules

These modules are added to the IG-XL workbook automatically. Do not edit these modules.

- Demod object

You use Demod properties and methods to create, define, and apply Demod instances that control the number and behavior of individual VSA instances, the individual VSA software processes.

The diagram highlights the first, and in this example, only Demod.Instance statement, shown in the OnProgramValidated interpose function in Exec_IP_Module. You create, define, and apply Demod instances in this interpose function because the VSA software takes about 15 seconds to initialize new instances and configure them using your definitions. Typically, you need a separate Demod instance for each different demodulator definition in your program. For best throughput, you create, define, and apply all instances in OnProgramValidated, or in a function called from that interpose function.

In the diagram, the arrow from the Demod object to the Demod service represents how the Demod instances control the VSA instances through the Demod service.

Demod objects are typically named for the wireless standard being tested, represented in the VBT statement by the string "STD".

The Demod object cannot be used in a DSP module.

- Source Signal Definition File (TSD file)

You use the `SignalDefinitionFile` property of the `Signals` interface of the `MWModulatedSource` to load a source signal definition. This `Signals` property specifies a TSD file to be loaded with the other `MWModulatedSource` `Signals` parameters. After the `LoadSignalsConnect` and the `Start` methods are executed, the waveform is sourced to the DUT by the instrument using the waveform samples and parameters in the TSD file (except for parameters that have been explicitly set to other values with additional `Signals` statements).

- [Measure Signal Definition File \(TSD File\)](#)

You use the `SignalDefinitionFile` property of the `Signals` interface of the `MWReceiver` to load a measure signal definition. As with the source, the `Signals` property specifies a TSD file to be loaded with the other `MWReceiver` `Signals` parameters. After the `LoadSignalsConnect` and the `Trigger` methods are executed, the DUT output signal is captured by the instrument using the parameters in the TSD file (except for parameters that have been explicitly set to other values with additional `Signals` statements).

Captured waveform samples are stored in a `DSPWave` object, as shown in the diagram.

- [DemodDSP](#) and `DemodDSPMacro` objects

The `DemodDSP` object allows multiple VSA instances to operate as demodulators in the IG-XL DSP environment. Up to two DSP computers with multiple core processors are available with an UltraFLEX test system. You use `DemodDSP` methods to create a VSA instruction queue which the Demod service uses to manage multiple VSA instances processing data for multiple Demod instances. The Demod service works with the DSP subsystem to allow multiple VSA instances processing capture data run in parallel in the background on the DSP computers. The Demod service uses `DemodDSP` methods and parameters in the instruction queue to control VSA instances using the .NET COM API.

You use a DSP procedure with a `DemodDSP` object for VSA demodulation processing in the same way as you normally use the DSP language and functions. You can run in any DSP mode and use deferred limits. The same restrictions apply, including no passing of strings.

The `DemodDSPMacro` object returns the results of a VSA macro to the `DemodDSP` object.

`DemodDSP` and `DemodDSPMacro` objects cannot be used in a VBT module, they must be in a DSP procedure in a DSP module. (Execution of `DemodDSP` and `DemodDSPMacro` is undefined outside of a DSP procedure.)

VSA

The embedded demodulator/analysis run-time software is a vector signal analyzer (Keysight Technologies 89600 VSA Software), or VSA. VSA accepts setup data and waveform data streams

under the control of the Demod service, providing the capability to demodulate and measure the characteristics of a broad variety of wireless standards-based signals.

VSA is shown in the lower right corner, communicating only with the Demod service. The Demod service communicates with the Demod, DemodDSP, and DemodDSPMacro objects, the Signals interfaces and IG-XL, to completely manage the VSA software running on the tester host and DSP computers. You use Demod instances in VBT modules and DemodDSP and DemodDSPMacro methods in DSP modules to control VSA instances (multiple, individual VSA process) through the Demod service. You use DemodDSP and DemodDSPMacro methods to specify VSA demodulation processing for a Demod instance and the related VSA instances.

The diagram shows multiple VSA instances under the top instance. In this case, these additional VSA instances are all defined by and associated with a single Demod instance. A program with multiple Demod instances can have many more VSA instances than shown. With the exception of a VSA instance for each Demod instance on the tester host (and two debugger instances as described below), they all run in the background on the DSP computers in the IG-XL DSP environment. The Demod service interacts directly with the IG-XL DSP environment, shown at the bottom of the diagram.

You need not independently launch and control VSA; the Demod service optimally manages all VSA instances. Avoid using VSA with your IG-XL test program independently of the Teradyne VBT objects and interfaces.

Debugger VSAs

Near VSA, in the lower right corner of the diagram, is a debugger VSA. When a program runs, and the first VSA instance is initialized by executing the first `Demod.Instances().ApplySettings` statement (that is, the first Demod instance definition is applied), the Demod service starts four additional VSA instances, called debugger instances, on the host computer. Like all other VSA instances, they are initially "invisible," with the VSA GUI turned off. But, unlike other VSA instances, you can turn on the VSA GUI (make it "visible") and link it to a specific VSA instance for a Demod instance to debug measurements using `Demod.Debuggers` properties and methods. You can open (make visible) and use up to two debugger VSAs at the same time.

A debugger VSA is linked to the input data and setup of a specific VSA instance, which can only be reconfigured with a different setup by changing the definition of the Demod instance for the VSA instance and reapplying the definition. But you can debug the demodulation measurement using the linked debugger VSA and then save the VSA setup for the measurement. You can then update the Demod instance definition with the new VSA setup.

Related Topics

- [Introduction to VSA](#)

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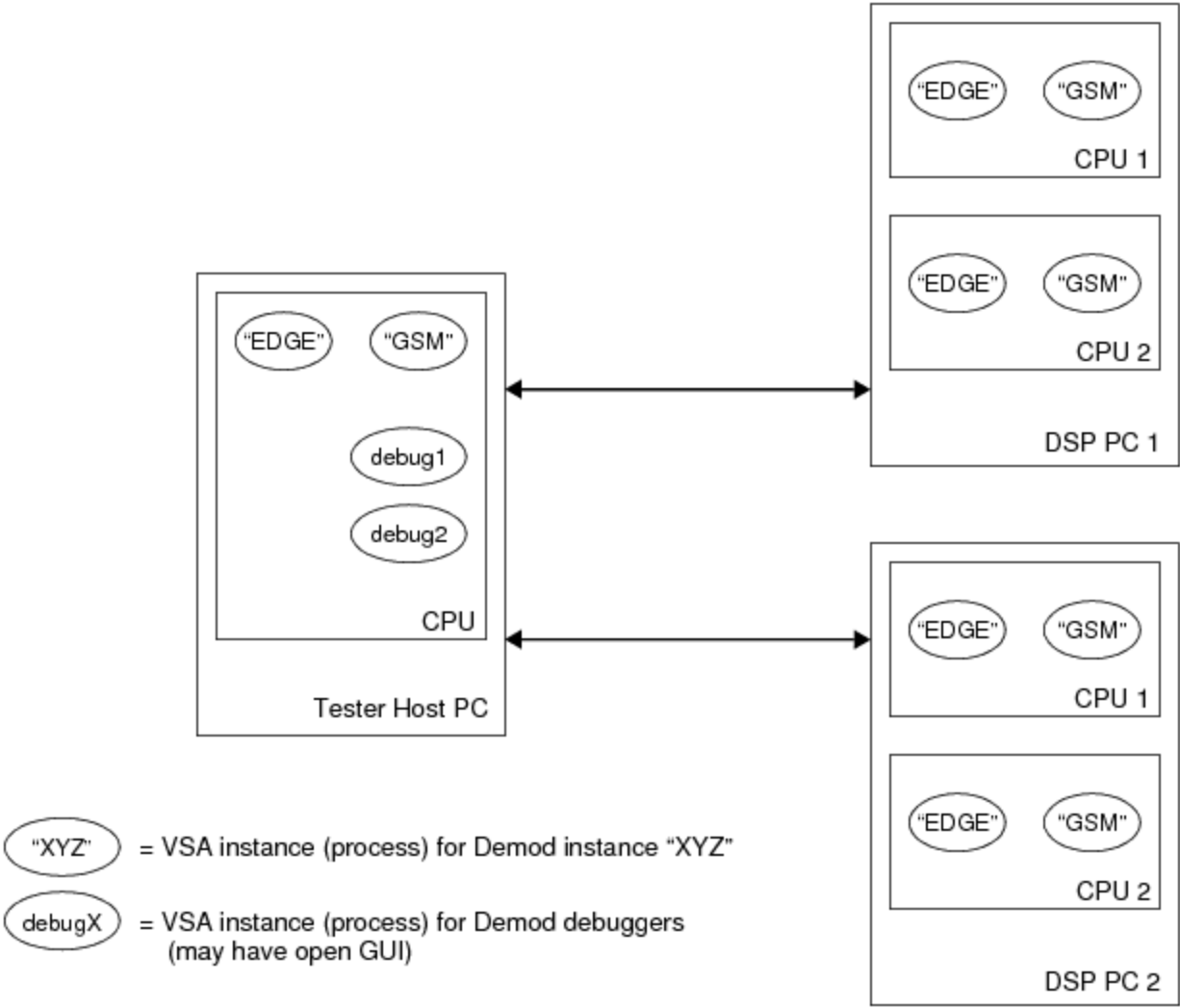
Using the ESA Toolkit : Demod Operation on Tester Computers

Demod Operation on Tester Computers

The demodulation features of the ESA Toolkit use multiple VSA processes on the host and DSP computers to demodulate and analyze data samples of modulated waveforms digitized by an UltraWave 12G or UltraWave24 receiver. The Demod service starts, initializes, and controls VSA processes depending on:

- The number Demod instances defined and applied in the test program
- The number of DSP PCs in the test system

The figure is a diagram of a test system with two dual-core DSP PCs.



Test System Computers and Demod VSA Processes

The diagram shows the VSA processes (instances) started by the Demod service and running on the test system PCs, after the test program defined and applied two Demod instances named EDGE and GSM. A VSA process (instance) copy is running for each Demod instance on each of the five CPUs in the system, and four debugger instances were started by the Demod service when the first Demod instance was applied.

The VSA instances on the host PC are used only in debug mode. In debug mode, the Demod service can link a debugger instance to a specific VSA instance for a site. A debugger instance may have an open (visible) GUI.

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Using the ESA Toolkit : Introduction to VSA

Introduction to VSA

The Vector Signal Analyzer (VSA) provided by the ESA Toolkit option is Keysight Technologies 89600 VSA Software, integrated into IG-XL during installation and workbook setup. Working with IG-XL, VSA can analyze and display modulated waveform data in TSD source signal files or in DSPWaves containing samples of DUT output waveforms captured by a receiver/digitizer such as an UltraWave 12G or UltraWave24 MWReceiver.

Keysight Technologies 89600 VSA Software provides the following key capabilities to the ESA Toolkit:

- Operation on digital I and Q input data or IF input data
- High resolution spectrum analysis
- Signal demodulators which perform analysis and measurements for major wireless standards

The ESA Toolkit software uses VSA as a compute engine for demodulation of standards-based RF signals. In the IG-XL DSP environment running on an UltraFLEX tester, a test program can run multiple copies of VSA in the background on DSP computers for increased throughput and parallel efficiency. Additionally, in debug mode, the ESA Toolkit uses VSA as a visible, GUI-based debugging tool.

Current wireless communications standards specify modulation schemes for various applications, such as cell phones, wireless networks, and wireless peripherals. Schemes such as QAM (quadrature amplitude modulation), QPSK (quadrature phase-shift keying), and PSK (phase-shift keying) are more complex than simple amplitude, frequency, or phase modulation. In addition, the characteristics of complex modulated signals change over a measurement cycle, as in burst, pulsed,

or TDMA (Time Division Multiple Access) formats. A signal is referred to as time-variant if it changes during the measurement period. Most digital communication signals are time-variant. Measuring these signals requires an instrument to capture the signal in a time period, limit the samples to the ones needed for a measurement, and measure the spectrum.

Vector signal analyzers provide the capabilities needed to make measurements on time-variant modulated signals. A complete RF vector signal analyzer:

- Downconverts RF signals to an IF or baseband signal
- Digitizes the IF or baseband signal while maintaining magnitude and phase data
- Transforms the sampled data into time and frequency domain information using DSP algorithms
- Computes wireless standard-specific information in the modulation and code domains using demodulator algorithms

VSA Licensing

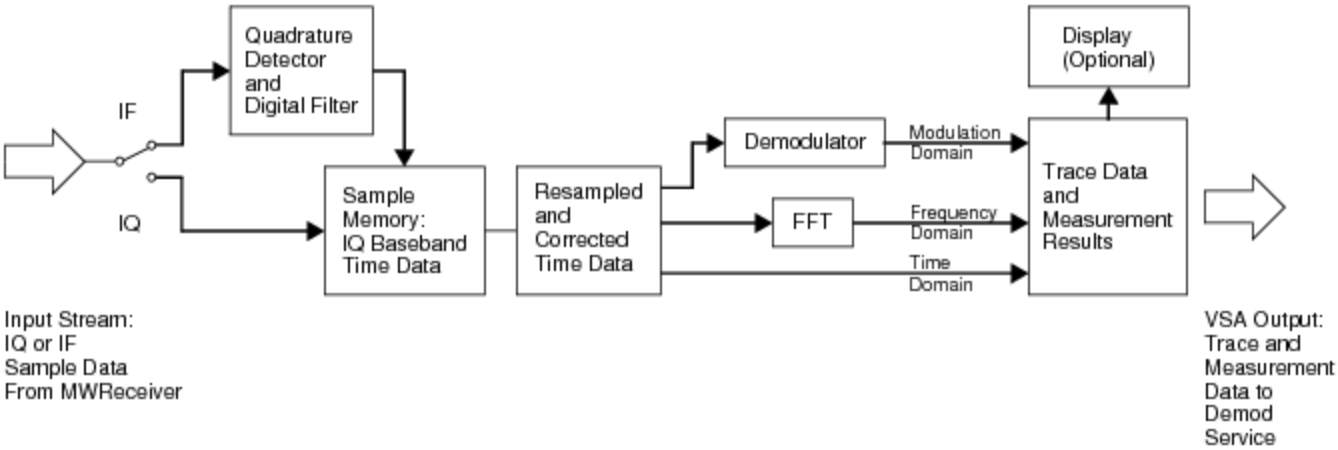
Keysight Technologies 89600 VSA Software in the ESA Toolkit requires one or more license options which you obtain directly from Teradyne. You need a license to enable ESA Toolkit, and you can also obtain licenses to work with waveforms for a variety of standards. For more information, see [ESA Toolkit Software Licensing Requirements](#).

VSA Operation with an UltraWave MWReceiver

As part of the ESA Toolkit, VSA software works with an UltraWave 12G or UltraWave24 MWReceiver instrument to function as an RF vector signal analyzer. The digitizer for the MWReceiver can send two kinds of data to the IG-XL DSP service in a DSPWave: IF data or IQ data output from the digital quadrature downconverter. In either case, the data is band-limited as required by the VSA.

The Demod service manages individual VSA processes (instances) and transfer of data to and from them. A VSA instance accepts input data in the form of one or more input streams managed by the Demod service.

The diagram shows the high-level operation of the VSA software, summarized from VSA Help. If the input stream is IF data, the VSA performs the IQ transformation in software.



VSA Software Block Diagram

A VSA demodulator requires minimal information to operate, such as modulation format, symbol rate, filter type and characteristics. This information is sent to a VSA instance in a “setup” when the instance is configured from the test program. The Demod service uses setup information from the VSA setup file referenced by the TSD file, and the Demod instance definition, to send the required setup to VSA.

For more information about VSA, see [VSA Help](#).

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Using the ESA Toolkit : Sourcing and Analyzing Modulated Signals

Sourcing and Analyzing Modulated Signals

The following topics describe programming concepts and procedures for specific programming tasks:

- Using Source and Capture Instruments
- Setting Up a Test Program for Demodulation Tests
- Creating and Defining Demod Instances
- Processing Data on VSA Instances
- Adding a Demodulation Measurement to a Program
- Extracting TSD Waveform Data to a Wave Definition

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Using the [ESA Toolkit : Sourcing and Analyzing Modulated Signals](#) : Using Source and Capture Instruments

Using Source and Capture Instruments

TSD files in the Teradyne Signal Library support an UltraWave 12G or UltraWave24 MWModulatedSource and MWReceiver. The VBT language and drivers for these instruments support referencing and loading a TSD file to set up non-DUT/application-specific parameters. You must set up DUT/application-specific parameters using Signals properties.

Other instruments that do not support TSD signals, such as BBAC and VHFAC, can extract waveform data into a WaveDefinition object. See [Extracting TSD Waveform Data to a Wave Definition](#). You must enter the TSD metadata manually using Signals properties.

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Using the ESA Toolkit : [Sourcing and Analyzing Modulated Signals](#) : Setting Up a Test Program for Demodulation Tests

Setting Up a Test Program for Demodulation Tests

After you install the ESA Toolkit, and before you use any demodulation components, you must set up the IG-XL workbook to use them.

To set up a workbook to use demodulation components:

1. [Type](#) Alt+F11 to start the Visual Basic editor.

2. [Select](#) File>Import File.

The Import File dialog appears.

3. [Navigate](#) to the bin folder where the IG-XL version is installed.

4. [Double-click](#) DSP_Declares_DemodDSP.bas.

The DSP_Declares_DemodDSP module appears in the workbook.

5. [Validate](#) the test program.

The DemodDSP module appears. Do not edit these two new modules.

6. [Save](#) the workbook.

The demodulation objects and libraries are now part of your workbook.

You can now use the Demod, DemodDSP, and DemodDSPMacro objects, measure TSD files, and VSA.

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Using the ESA Toolkit : Sourcing and Analyzing Modulated Signals : Creating and Defining Demod Instances

Creating and Defining Demod Instances

You must create individual, unique, Demod instances with definitions for each specific standard and type of VSA processing in a test program.

Initializing VSA instances takes about 15 seconds. For best throughput, create, define, and apply Demod instances in the OnProgramValidated interpose function.

To create and define a Demod instance:

- | | |
|----|--|
| 1. | Add a Demod instance with a unique name that identifies the type of demodulation processing. |
| 2. | Assign values for the definition properties. |
| 3. | Specify a VSA setup file. |
| 4. | Call the ApplySettings method. |

The Demod service creates the VSA instance, copies it to the DSP PCs, and initializes all VSA instances.

To change the setup of the VSA instances for a Demod instance, you must redefine the Demod instance and call ApplySettings again.

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Using the ESA Toolkit : Sourcing and Analyzing Modulated Signals : Processing Data on VSA Instances

Processing Data on VSA Instances

Running VSA instances are a “pool” of compute engines. Copies of the same VSA instance run on each core of each DSP PC, plus a copy runs on the host computer for host debug mode. To process data and produce results, VSA requires the following steps:

1. Input the data.
2. Optionally, run one or more macros.
3. Read one or more results from VSA tables/markers.
4. Optionally, read multi-element trace data.

All steps must be run on the same specific copy of a VSA instance to get correct results. To manage the pool of VSA instances, the Demod Service uses a VSA instruction queue. The queue is a group of processing steps that must be executed together to get correct results. The Demod service optimizes use of instances for maximum throughput. Create a VSA instruction queue for each of the Demod instances you create and define. To process data for a Demod instance on multiple VSA instances:

1. Create a a DSP procedure in the DSP module, typically DSP_Module.

A DSP procedure may have more than one instruction queue.

Do not use DSP methods on the DSPWave containing the waveform samples between instructions in the queue.

2. Create a VSA instruction queue using a sequence of DemodDSP methods.

Methods in the queue all require the name of the Demod instance as a parameter. Since you cannot pass a string into a DSP procedure, specify the instance name as a string literal.

- a. Start each queue with InitProcessing().
- b. Assign the waveform data in a DSPWave to a VSA input using AssignInputData().
- c. Add one or more methods to specify the measurement result you want.

Typically, you use AssignTableValue() or AssignResult(). The first method is a lower level call to VSA and requires some knowledge of the VSA COM API. (Note that the AssignResult method will not have any additional enumerations added in the IG-XL 8.10/ESA Toolkit 3.5 release and later.)

If you call more than one measurement method with the same result variable, only the last measurement result is returned. The previous measurement results are lost.

- d. End the queue using ExecuteProcessing().

None of the statements in the queue execute until the end of the queue.

The instruction queue can also include calls to VSA macros or to return VSA trace data in a DSPWave. At the end of the queue, the Demod service executes the entire queue using one or more VSA instances for defined Demod instance.

Example

Set up a VSA instruction queue to control the processing of the VSA instances set up by the Demod instance named I802_11a_Demod to make an EVM measurement:

' DSP procedure in a DSP Module

```
Public Function I80211aEvm(ByVal IfCaptureData As DspWave, _
    ByRef EvmRmsPercent As Double) As Long
```

```
Dim Status As Long
```

' Define instruction queue - instance name is a string literal

```
Status = DemodDSP.InitProcessing("I802_11a_Demod")
```

```
Status = DemodDSP.AssignInputData("I802_11a_Demod", IfCaptureData, 1)
```

```
Status = DemodDSP.AssignResult("I802_11a_Demod", tIDemod80211n1x1EvmRms, _
    EvmRmsPercent)
```

' Process data and get result

Status = DemodDSP.ExecuteProcessing("I802_11a_Demod", True)

End Function

	
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Using the ESA Toolkit : Sourcing and Analyzing Modulated Signals : Adding a Demodulation Measurement to a Program

Adding a Demodulation Measurement to a Program

This procedure assumes that you have an IG-XL workbook set up with the demodulation tools (see [Setting Up a Test Program for Demodulation Tests](#)) and that you have used SDE to copy the measure TSD file to a folder accessible from the workbook folder.

To create or modify a test program to capture and analyze a wireless standards-based modulated signal:

1. [Create and define a Demod instance](#) for your measurement. See [Creating and Defining Demod Instances](#).

This Demod instance will reference the measure TSD file, which includes a VSA setup file.

2. [Instantiate the VSA instances using the ApplySettings method](#) for the Demod instance in OnProgramValidated().

[Applying the definition to each](#) Demod instance takes about 15 seconds. For best throughput, instantiate VSA instances before the tests run in the program.

3. [Create a capture Signal](#).

- a. [Find the place in the VBT code where the Signals are defined for the capture instrument](#).

- b. [Add a Signal for the capture instrument and then add parameters to define the Signal using the a TSD file](#).

This defines the subset of Signals parameters defined by the TSD file properties.

- c.

Define the remaining Signals parameters, such as expected power and center frequency, and any others.
4. Set up the capture instrument.

a.

Find the place in the VBT code where data is to be captured.

b.

Set up the capture instrument using the Signal defined above.

c.

Trigger a capture using VBT or a pattern microcode.
5. Add a DSP Procedure in a DSP module.
6. Add a call to a DSP procedure and pass the DSPWave with the captured data to a VSA instruction queue.
7. Add an instruction queue to the DSP procedure to define the VSA processing for the Demod instance. See Processing Data on VSA Instances.
8. Validate and debug the test program.

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Using the [ESA Toolkit : Sourcing and Analyzing Modulated Signals](#) : Extracting TSD Waveform Data to a Wave Definition

Extracting TSD Waveform Data to a Wave Definition

You can export the waveform data in a source TSD file to a wave definition and then use the wave definition with an instrument that does not have Signals interface support for importing a signal definition in a TSD file. For example, you can extract the separate I and Q waveform data for a TSD file and use it to create separate I and Q signals with the BBACSrc.

The [IG-XL method](#) `TheExec.WaveDefinitions.CreateWaveDefinitionFromFile` allows you to perform this extraction. See [CreateWaveDefinitionFromFile](#).

After you extract the waveform data from the TSD file, you must manually transfer some Signals parameters, such as `SampleRate` and `SampleSize`, from the TSD file properties.

Any wave definition you create in this way is not be available for reference on a Wave Definitions sheet or within WaveDesigner tool (compound definitions, for example).

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Using the ESA Toolkit : Debugging Demodulation Tests

Debugging Demodulation Tests

The Demod service starts two debugger VSA instances the first time a Demod instance definition is applied. You can link and open (start the VSA GUI) one or more debuggers using `Demod.Debuggers` properties and methods.

The debugger VSA opens using the current setup of the VSA instance. The Demod service redirects the input data from the VSA instance to the debugger.

Debuggers link to the copy of the user instance running on the host. You must run in host debug mode or step into a DSP procedure and execute on the host to debug the DSP procedure code.

How the Debugger VSA and VSA Instances Interact

You can change the configuration of the debugger VSA using the GUI or VBT. However, GUI controls on a debugger window or VBT statements do not affect VSA instances. To change the setup of a VSA instance, you must save a setup file and reapply the updated definition to the Demod instance.

When you debug your test program, all instructions in the VSA instance queue are also executed on the linked debugger.

During step-by-step debugging, each instruction is executed immediately on the debugger instead of waiting for `ExecuteProcessing()`. This allows you to see the effects of the execution of each DemodDSP method on VSA. Results from a VSA instance are unavailable until after the end of the instruction queue.

You can change a VSA instance using VBT statements, but this does not change the linked debugger. Use debugger language to update the debugger VSA.

Results returned to the DSP procedure are always the results from the VSA instance and not from the linked debugger. A VSA debugger with a modified configuration may show different results.

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